PPN-1: Parameterized Post-Newtonian Parameters in Spectral Causal Theory

Three Levels of Approximation from the NT-4a Propagator

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Abstract

We derive the Parameterized Post-Newtonian (PPN) parameters of Spectral Causal Theory (SCT) at three levels of approximation. **Level 1** (exact nonlocal) formulates the gravitational potentials $\Phi(r)$ and $\Psi(r)$ as Fourier–Bessel integrals over the full SCT propagator denominators $\Pi_{\mathrm{TT}}(z)$ and $\Pi_{\mathrm{s}}(z,\xi)$, and identifies the structural obstructions to naive numerical evaluation (nonzero UV asymptote, real-axis pole, prescription mismatch). **Level 2** (local Yukawa) replaces the propagator denominators by their leading pole approximants, yielding analytic Yukawa potentials with two effective masses $m_2 = \Lambda\sqrt{60/13}$ and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$, from which the linear PPN parameter $\gamma(r)$ and lower bounds on Λ are extracted. **Level 3** (nonlinear PPN) identifies the remaining parameters (β, α_i, ζ_i, ξ_{PPN}) as requiring the $\mathcal{O}(h^2)$ field equations from NT-4b, and marks them as not yet derived.

Within the Level 2 approximation, the Cassini bound gives $\Lambda \gtrsim 6.3 \times 10^{-18}\,\mathrm{eV}$, the MESSENGER bound gives $\Lambda \gtrsim 6.3 \times 10^{-18}\,\mathrm{eV}$, and the Eot-Wash torsion-balance bound gives $\Lambda \gtrsim 2.4 \times 10^{-3}\,\mathrm{eV}$. The MICROSCOPE and Lunar Laser Ranging bounds require β and are deferred.

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1 Scope and Conventions

1.1 Purpose and Structure

The purpose of PPN-1 is to connect the verified spectral form factors and the NT-4a propagator with observable weak-field quantities. The derivation proceeds in three nested levels of approximation:

Level 1: **Exact nonlocal.** The gravitational potentials are written as Fourier–Bessel integrals involving the full propagator denominators. This is mathematically exact (within the linearized static sector) but not yet numerically implemented in a reliable way due to a real-axis pole and a nonzero UV asymptote in the momentum-space kernels.

Level 2: **Local Yukawa.** The propagator denominators are replaced by their leading simple-pole approximants, reducing the potentials to Stelle-like Yukawa form. This is the operative approximation for all solar-system phenomenology and is fully verified.

Level 3: **Nonlinear PPN.** The $\mathcal{O}(h^2)$ terms in g_{00}, the g_{0i} gravitomagnetic sector, and the nonlinear superposition structure are needed for the remaining PPN parameters (β, α_i, ζ_i, $\xi_{\rm PPN}$). These require the nonlinear field equations from NT-4b and are not derived here.

The following quantities are *not* claimed in this document:

a) a full ten-parameter PPN completion,

b) a derivation of β,

c) a proof that the preferred-frame or conservation parameters vanish,

d) a reliable numerical evaluation of the exact nonlocal Fourier integrals at all radii.

1.2 Metric and Sign Conventions

We use the Lorentzian weak-field form

$$g_{00} = -1 + 2\Phi(r), \qquad g_{ij} = \big(1 + 2\Psi(r)\big)\delta_{ij}, \tag{1}$$

with signature $(-,+,+,+)$. In the static limit, the physical potential energy of a non-relativistic test body is

$$V(r) = -m_{\rm test}\,\Phi(r). \tag{2}$$

The Newtonian limit corresponds to

$$\Phi_{\rm N}(r) = \Psi_{\rm N}(r) = \frac{GM}{r}, \qquad V_{\rm N}(r) = -\frac{GMm_{\rm test}}{r}. \tag{3}$$

1.3 Dimensional Conventions

Natural units $c = \hbar = 1$ are used throughout. For phenomenology we restore

$$\hbar c = 1.9732705 \times 10^{-7}\,\mathrm{eV}\cdot\mathrm{m}, \qquad 1\,\mathrm{m} \simeq 5.06773 \times 10^{6}\,\mathrm{eV}^{-1}. \tag{4}$$

Key length scales:

$$1\,\mathrm{AU} \simeq 1.496 \times 10^{11}\,\mathrm{m} \simeq 7.58 \times 10^{17}\,\mathrm{eV}^{-1}, \tag{5}$$

$$38.6\,\mu\mathrm{m} \simeq 1.96 \times 10^{2}\,\mathrm{eV}^{-1}. \tag{6}$$

2 Input from NT-1b Phase 3 and NT-4a

The starting point is the verified one-loop spectral action in the Weyl basis,

$$\Gamma^{(1)} = \frac{1}{16\pi^2} \int \mathrm{d}^4x \sqrt{g} \left[\alpha_C \hat{F}_1\left(\frac{\Box}{\Lambda^2}\right) C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} + \alpha_R(\xi) \hat{F}_2\left(\frac{\Box}{\Lambda^2}\right) R^2 \right], \tag{7}$$

with the verified coefficients

$$\alpha_C = \frac{13}{120}, \qquad \alpha_R(\xi) = 2\left(\xi - \frac{1}{6}\right)^2, \tag{8}$$

and normalized form factors

$$\hat{F}_i(z) := \frac{F_i(z)}{F_i(0)}, \qquad \hat{F}_i(0) = 1. \tag{9}$$

The SM field content is $N_s = 4$ real scalars, $N_D = 45/2$ Dirac fermions, $N_v = 12$ gauge bosons (CPR 0805.2909).

The NT-4a propagator denominators are

$$\Pi_{\mathrm{TT}}(z) = 1 + c_2\, z\, \hat{F}_1(z), \qquad c_2 = 2\alpha_C = \frac{13}{60}, \tag{10}$$

$$\Pi_{\mathrm{s}}(z,\xi) = 1 + (3c_1 + c_2)\, z\, \hat{F}_2(z,\xi) = 1 + 6\left(\xi - \frac{1}{6}\right)^2 z\, \hat{F}_2(z,\xi). \tag{11}$$

Two structural facts matter:

1. $\Pi_{\mathrm{TT}}(0) = 1$ and $\Pi_{\mathrm{s}}(0,\xi) = 1$, so the infrared limit is Einsteinian.
2. At conformal coupling $\xi = 1/6$, one has $\Pi_{\mathrm{s}}(z, 1/6) \equiv 1$, so the scalar channel decouples from the linearized dynamics.

3 Level 1: Exact Linear Static Potentials

3.1 From Linearized Field Equations to Fourier Representation

Consider a static, spherically symmetric point source:

$$T_{00} = M\, \delta^{(3)}(\vec{x}), \qquad T_{0i} = T_{ij} = 0. \tag{12}$$

The linearized SCT field equations (NT-4a) in Fourier space, after decomposition into spin-2 (transverse-traceless) and spin-0 (scalar trace) sectors via Barnes–Rivers projectors $P^{(2)}$ and $P^{(0\text{-}s)}$, take the form

$$\Pi_{\mathrm{TT}}(k^2/\Lambda^2)\, \tilde{h}^{\mathrm{TT}}_{\mu\nu}(k) = -\frac{16\pi G}{k^2}\, P^{(2)}_{\mu\nu,\rho\sigma}\, \tilde{T}^{\rho\sigma}(k), \tag{13}$$

and similarly for the scalar sector with Π_{s} and $P^{(0\text{-}s)}$. The metric perturbation in position space is obtained by inverse Fourier transform.

For the static, isotropic metric (1), the two potentials Φ and Ψ are extracted by projecting onto g_{00} and $g_{ij} \propto \delta_{ij}$ components. The spin-2 mode contributes to both Φ and Ψ with definite relative weights (the van Dam–Veltman factor of 4/3 for spin-2 vs. spin-0 in the temporal potential), while the spin-0 mode contributes with $-1/3$ to Φ and $+1/3$ to Ψ. This yields the Newton kernels

$$K_\Phi(z) = \frac{4}{3\,\Pi_{\mathrm{TT}}(z)} - \frac{1}{3\,\Pi_{\mathrm{s}}(z,\xi)}, \tag{14}$$

$$K_\Psi(z) = \frac{2}{3\,\Pi_{\mathrm{TT}}(z)} + \frac{1}{3\,\Pi_{\mathrm{s}}(z,\xi)}, \tag{15}$$

where $z = k^2/\Lambda^2$. At $z = 0$: $K_\Phi(0) = 4/3 - 1/3 = 1$ and $K_\Psi(0) = 2/3 + 1/3 = 1$, recovering the Newtonian limit $\Phi = \Psi = GM/r$.

3.2 Angular Averaging: From 3D Fourier to Fourier–Bessel

The position-space potential is

$$\Phi(r) = \frac{GM}{(2\pi)^3} \int \mathrm{d}^3k \, \frac{e^{i\vec{k}\cdot\vec{r}}}{k^2} \, K_\Phi(k^2/\Lambda^2). \tag{16}$$

In spherical coordinates (k, θ, φ) with $\vec{k}\cdot\vec{r} = kr\cos\theta$, the angular integral gives

$$\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \mathrm{d}\theta \, \sin\theta \, e^{ikr\cos\theta} = 2\pi \cdot \frac{2\sin(kr)}{kr} = \frac{4\pi\sin(kr)}{kr}. \tag{17}$$

Therefore

$$\Phi(r) = \frac{GM}{(2\pi)^3} \int_0^\infty 4\pi k^2 \mathrm{d}k \, \frac{1}{k^2} \, \frac{4\pi\sin(kr)}{kr} \, K_\Phi(k^2/\Lambda^2) = \frac{GM}{r} \, \frac{2}{\pi} \int_0^\infty \frac{\sin(kr)}{k} \, K_\Phi(k^2/\Lambda^2) \, \mathrm{d}k. \tag{18}$$

The factor GM/r has been separated using the standard Dirichlet integral

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kr)}{k} \, \mathrm{d}k = 1, \qquad r > 0. \tag{19}$$

Proposition 3.1 (Exact linear static potentials)**.** *For the point source* (12)*, the linearized SCT potentials are*

$$\frac{\Phi(r)}{\Phi_{\mathrm{N}}(r)} = \frac{2}{\pi} \int_0^\infty \frac{\sin(kr)}{k} \, K_\Phi(k^2/\Lambda^2) \, \mathrm{d}k, \tag{20}$$

$$\frac{\Psi(r)}{\Psi_{\mathrm{N}}(r)} = \frac{2}{\pi} \int_0^\infty \frac{\sin(kr)}{k} \, K_\Psi(k^2/\Lambda^2) \, \mathrm{d}k. \tag{21}$$

These are exact within the linearized static sector.

3.3 UV Behavior of the Newton Kernels

Unlike the Stelle (local higher-derivative) case, where $\Pi_{\mathrm{TT}}(z) = 1 + c_2 z$ grows linearly and hence $K_\Phi(z) \to 0$ as $z \to \infty$, the SCT propagator denominators involve the entire form factor $\hat{F}_1(z)$ which has a nontrivial large-z asymptotic. Specifically, the NT-1b Phase 3 result gives

$$\Pi_{\mathrm{TT}}(z) \xrightarrow{z\to\infty} 1 + c_2 \, z \, \hat{F}_1(z) \;\to\; c_2 \left(-\tfrac{89}{12}\right) \cdot \frac{1}{\alpha_C} = -\frac{89}{12} \cdot \frac{F_1(\infty)}{F_1(0)} \cdot c_2 \, z \to \cdots \tag{22}$$

Numerically, $\Pi_{\mathrm{TT}}(z)$ saturates to a finite negative value:

$$\Pi_{\mathrm{TT}}(z \to \infty) \approx -13.83. \tag{23}$$

This means the Newton kernel has a nonzero UV asymptote:

$$K_\Phi(z \to \infty) = \frac{4}{3 \cdot (-13.83)} - \frac{1}{3 \cdot \Pi_{\mathrm{s}}(\infty)} \approx -0.136. \tag{24}$$

Consequently, the correction integral

$$\int_0^\infty \frac{\sin(kr)}{k} \big[K_\Phi(z) - 1\big] \mathrm{d}k$$

converges only conditionally through the oscillatory $\sin(kr)$ factor, not absolutely. This is a fundamental difference from Stelle gravity where $K_\Phi - 1 \to -1$ at $z \to \infty$, making the integral absolutely convergent after standard Yukawa decomposition.

3.4 The TT Pole on the Real Axis

The propagator denominator $\Pi_{\mathrm{TT}}(z)$ has a zero on the positive real axis at

$$z_0 \approx 2.41484\ldots, \qquad \Pi_{\mathrm{TT}}(z_0) = 0. \tag{25}$$

This zero produces a simple pole in $1/\Pi_{\mathrm{TT}}(z)$, and hence in $K_\Phi(z)$ and $K_\Psi(z)$, at $z = z_0$. In momentum space, this corresponds to a pole at $k = k_0 := \Lambda\sqrt{z_0}$ on the real integration axis of the Fourier–Bessel integrals (20)–(21).

The residue of the pole in $1/\Pi_{\mathrm{TT}}(z)$ at $z = z_0$ is

$$R_{\mathrm{pole}} = \frac{1}{\Pi'_{\mathrm{TT}}(z_0)} \approx -1.191, \tag{26}$$

where $\Pi'_{\mathrm{TT}}(z_0) \approx -0.8398$.

Remark 3.2 (Comparison with Stelle gravity)**.** *In Stelle's local theory,* $\Pi_{\mathrm{TT}}(z) = 1 + c_2 z$ *has its zero at* $z_{\mathrm{Stelle}} = -1/c_2 = -60/13 \approx -4.615$*, which lies on the* negative *real axis. Since* $z = k^2/\Lambda^2 \geq 0$ *on the integration domain, the Stelle pole is at imaginary* k *and produces a standard Yukawa term* $e^{-m_2 r}$ *without entering the integration contour. In SCT, the zero at* $z_0 \approx 2.41 > 0$ *lies on the integration domain, creating a qualitatively different analytic structure.*

3.5 Prescription Issue for the Pole Contribution

The pole at $k = k_0$ on the real integration axis requires a prescription. Two natural choices exist:

Principal value (PV). Subtracting the pole on the real axis and adding back the PV integral analytically. The Gradshteyn–Ryzhik identity (3.723.2) for the real-axis pole gives

$$\frac{2}{\pi}\,\mathrm{P.V.}\!\int_0^\infty \frac{\sin(kr)}{k}\,\frac{\Lambda^2}{k^2 - k_0^2}\,\mathrm{d}k = \frac{1-\cos(k_0 r)}{k_0^2/\Lambda^2}, \tag{27}$$

producing an *oscillatory* contribution $\propto \cos(k_0 r)$.

Euclidean prescription. The spectral action is defined via the Euclidean path integral, where $k^2 \to -k_E^2$, and the pole in $1/\Pi_{\mathrm{TT}}$ contributes as

$$\frac{2}{\pi}\int_0^\infty \frac{\sin(kr)}{k}\,\frac{k_0^2}{k^2 + k_0^2}\,\mathrm{d}k = \frac{1 - e^{-k_0 r}}{1}, \tag{28}$$

producing a *Yukawa-damped* contribution $\propto e^{-k_0 r}$.

Remark 3.3 (Prescription mismatch)**.** *If the regularized correction integral (with the pole subtracted on the real* k*-axis) is combined with the Euclidean pole contribution, the two pieces are* inconsistent*: the subtracted PV integral removes the cosine behavior while the addback introduces exponential damping, creating a mismatch of magnitude*

$$\Delta = \frac{4}{3}\,\frac{R_{\mathrm{pole}}}{z_0}\,\left[2 - e^{-k_0 r} - \cos(k_0 r)\right]. \tag{29}$$

A numerically correct treatment requires either (a) direct Euclidean-space integration, (b) Levin-type oscillatory quadrature with analytic continuation, or (c) contour deformation in the complex k*-plane. These methods are deferred to a dedicated numerical study.*

3.6 Current Status of Level 1

The Level 1 formulation is mathematically exact and well-defined. However, the numerical implementation of the Fourier–Bessel integrals is not yet reliable due to the real-axis pole (section 3.4) and the nonzero UV asymptote (section 3.3). The functions `phi_exact`, `psi_exact`, `gamma_exact` in `ppn1_parameters.py` document the mathematical structure but are marked as PLACEHOLDER. For all quantitative phenomenology, Level 2 applies.

Remark 3.4 (Validity of Level 2 for solar-system tests)**.** *For any viable Λ (i.e., Λ satisfying the experimental bounds below), we have $r_{\rm AU} \cdot m_2 \gg 1$. In this regime, the Yukawa corrections are exponentially suppressed, and the difference between the exact nonlocal potentials and the local Yukawa approximation is smaller than the current experimental precision. Level 2 is therefore the correct approximation for all solar-system observables at present.*

4 Level 2: Local Yukawa Approximation

4.1 Small-z Expansion and Effective Masses

For $k^2 \ll \Lambda^2$ (i.e., $z \ll 1$), the normalized form factors satisfy $\hat{F}_i(z) = 1 + \mathcal{O}(z)$, so

$$\Pi_{\rm TT}(z) = 1 + c_2\, z + \mathcal{O}(z^2), \qquad \Pi_{\rm s}(z,\xi) = 1 + (3c_1 + c_2)\, z + \mathcal{O}(z^2). \tag{30}$$

The denominators have the same form as Stelle gravity with effective masses

$$m_2^2 = \frac{\Lambda^2}{c_2} = \frac{60}{13}\Lambda^2, \qquad m_2 = \Lambda\sqrt{\frac{60}{13}} \approx 2.1483\,\Lambda, \tag{31}$$

$$m_0^2 = \frac{\Lambda^2}{3c_1 + c_2} = \frac{\Lambda^2}{6(\xi - 1/6)^2}, \qquad m_0(\xi = 0) = \Lambda\sqrt{6} \approx 2.4495\,\Lambda. \tag{32}$$

At exact conformal coupling $\xi = 1/6$, the scalar mode decouples and $m_0 = \infty$.

4.2 Local Yukawa Potentials via Fourier–Bessel Integral

In the local approximation, the propagator denominators become simple polynomials: $\Pi_{\rm TT}(z) \approx 1 + c_2 z$ and $\Pi_{\rm s}(z,\xi) \approx 1 + (3c_1 + c_2)z$. The Newton kernels decompose into partial fractions:

$$K_\Phi(z) = \frac{4}{3}\frac{1}{1 + c_2 z} - \frac{1}{3}\frac{1}{1 + (3c_1 + c_2)z} = 1 + \frac{4}{3}\frac{m_2^2}{k^2 + m_2^2}\cdot(-1) - \frac{1}{3}\frac{m_0^2}{k^2 + m_0^2}\cdot(-1) + \cdots \tag{33}$$

More precisely, using $1/(1 + c_2 z) = m_2^2/(k^2 + m_2^2)$, we have

$$K_\Phi^{\rm loc}(z) = \frac{4}{3}\frac{m_2^2}{k^2 + m_2^2} - \frac{1}{3}\frac{m_0^2}{k^2 + m_0^2}. \tag{34}$$

The Fourier–Bessel identity (Gradshteyn–Ryzhik 3.723.2) states

$$\frac{2}{\pi}\int_0^\infty \frac{\sin(kr)}{k}\frac{m^2}{k^2 + m^2}\,{\rm d}k = 1 - e^{-mr}, \qquad r > 0,\ m > 0. \tag{35}$$

Applying this to each term:

Proposition 4.1 (Local Yukawa potentials)**.** *In the local Yukawa approximation, for $\xi \neq 1/6$,*

$$\boxed{\frac{\Phi_{\rm loc}(r)}{\Phi_{\rm N}(r)} = 1 - \frac{4}{3}\,e^{-m_2 r} + \frac{1}{3}\,e^{-m_0 r},} \tag{36}$$

$$\boxed{\frac{\Psi_{\rm loc}(r)}{\Psi_{\rm N}(r)} = 1 - \frac{2}{3}\,e^{-m_2 r} - \frac{1}{3}\,e^{-m_0 r}.} \tag{37}$$

At conformal coupling ($\xi = 1/6$, $m_0 = \infty$), the scalar terms are absent:

$$\frac{\Phi_{\rm loc}(r)}{\Phi_{\rm N}(r)}\bigg|_{\xi=1/6} = 1 - \frac{4}{3}\,e^{-m_2 r}, \qquad \frac{\Psi_{\rm loc}(r)}{\Psi_{\rm N}(r)}\bigg|_{\xi=1/6} = 1 - \frac{2}{3}\,e^{-m_2 r}. \tag{38}$$

Proof. Substitute $K_\Phi^{\rm loc}$ and $K_\Psi^{\rm loc}$ into (20)–(21) and apply the identity (35) term by term. Note that $4/3 \cdot 1 - 1/3 \cdot 1 = 1$ and $2/3 \cdot 1 + 1/3 \cdot 1 = 1$, confirming the GR limit at large r. □

Remark 4.2. *The coefficients 4/3 and 2/3 for the spin-2 Yukawa, and ±1/3 for the scalar Yukawa, are fixed by the spin decomposition of the graviton propagator. They are universal for all higher-derivative gravity theories with the Weyl^2 + R^2 structure, and agree with Stelle (1977) and Edholm, Koshelev, Mazumdar (2016).*

4.3 Short-Distance Structure

Proposition 4.3 (Short-distance limits)**.** *For $\xi \neq 1/6$, the local potentials are regular at the origin:*

$$\Phi_{\rm loc}(r) = GM\left(\frac{4}{3}\,m_2 - \frac{1}{3}\,m_0\right) + \mathcal{O}(r), \qquad r \to 0. \tag{39}$$

At exact conformal coupling,

$$\Phi_{\rm loc}(r)\big|_{\xi=1/6} \sim -\frac{GM}{3r}, \qquad r \to 0. \tag{40}$$

Proof. Expand $e^{-mr} = 1 - mr + \mathcal{O}(r^2)$ in eqs. (36) and (37). The coefficient of $1/r$ is $1 - 4/3 + 1/3 = 0$ when $\xi \neq 1/6$. At conformal coupling, the scalar term is absent and $1 - 4/3 = -1/3$. □

Remark 4.4. *The conformal and short-distance limits do not commute: taking $\xi \to 1/6$ at fixed $r > 0$ removes the scalar Yukawa tail, whereas taking $r \to 0$ first at non-conformal ξ shows the cancellation of the Newtonian singularity.*

5 The Linear PPN Parameter $\gamma(r)$

5.1 Definition and Derivation

The linear PPN parameter is defined as

$$\gamma(r) := \frac{\Psi(r)}{\Phi(r)}. \tag{41}$$

In GR, $\gamma = 1$ identically. Within the Level 2 approximation:

Proposition 5.1 (Level 2 gamma)**.**

$$\boxed{\gamma_{\rm loc}(r) = \frac{1 - \frac{2}{3}\,e^{-m_2 r} - \frac{1}{3}\,e^{-m_0 r}}{1 - \frac{4}{3}\,e^{-m_2 r} + \frac{1}{3}\,e^{-m_0 r}}.} \tag{42}$$

5.2 Large-Distance (GR) Limit

Proposition 5.2 (GR recovery)**.** *For $r \gg 1/m_2$,*

$$\gamma_{\rm loc}(r) = 1 + \mathcal{O}(e^{-m_2 r}, e^{-m_0 r}). \tag{43}$$

Proof. Both exponentials vanish at large radius; numerator and denominator of eq. (42) both approach unity. □

More precisely, for $m_0 \geq m_2$ (which holds for all ξ since $m_0/m_2 = \sqrt{60/(78(\xi - 1/6)^2)} \geq 1$ in the relevant parameter range), the leading correction is

$$\gamma(r) - 1 \approx \frac{2}{3}\, e^{-m_2 r} \qquad (r \gg 1/m_2). \tag{44}$$

5.3 Short-Distance Values

For $\xi \neq 1/6$, taking $r \to 0$:

$$\gamma_{\rm loc}(0) = \frac{2m_2 + m_0}{4m_2 - m_0}. \tag{45}$$

Numerical values for representative ξ:

ξ	m_0/Λ	m_2/Λ	$\gamma(0)$
0	$\sqrt{6} \approx 2.449$	$\sqrt{60/13} \approx 2.148$	≈ 1.098
1/6 (conformal)	∞	2.148	-1

At exact conformal coupling, $\gamma(0) = (2/3)/(-1/3) = -1$, reflecting the absence of the scalar mode that would otherwise cancel the Newtonian singularity.

6 Phenomenological Bounds on Λ

In this section, we extract lower bounds on the SCT scale Λ from five experiments. The first three (Cassini, MESSENGER, Eot-Wash) constrain γ and yield direct bounds. The last two (MICROSCOPE, LLR) constrain the Nordtvedt parameter $\eta = 4\beta - \gamma - 3$ and require knowledge of β, which is not yet derived.

6.1 Cassini (Shapiro Time Delay)

The Cassini spacecraft measurement of the Shapiro time delay (Bertotti, Iess, Tortora 2003) constrains

$$|\gamma - 1| < 2.3 \times 10^{-5} \tag{46}$$

at a distance $r \simeq 1\,\text{AU}$. Using the asymptotic form (44):

$$\frac{2}{3}\, e^{-m_2 r_{\rm AU}} < 2.3 \times 10^{-5}, \qquad \Longrightarrow \qquad e^{-m_2 r_{\rm AU}} < 3.45 \times 10^{-5}. \tag{47}$$

Taking logarithms:

$$m_2\, r_{\rm AU} > -\ln(3.45 \times 10^{-5}) \approx 10.274. \tag{48}$$

With $r_{\rm AU} \simeq 7.58 \times 10^{17}\,\text{eV}^{-1}$ and $m_2 = \Lambda\sqrt{60/13}$:

$$\boxed{\Lambda \gtrsim 6.31 \times 10^{-18}\,\text{eV} \qquad \text{(Cassini, Level 2 Yukawa, spin-2 dominant).}} \tag{49}$$

6.2 MESSENGER (Radioscience)

The MESSENGER radioscience analysis (Verma, Fienga, Laskar+ 2014) provides a tighter constraint on γ:

$$|\gamma - 1| < 2.0 \times 10^{-5}. \tag{50}$$

The same analysis gives

$$e^{-m_2 r_{\rm AU}} < 3.0 \times 10^{-5}, \qquad m_2\, r_{\rm AU} > 10.41. \tag{51}$$

$$\boxed{\Lambda \gtrsim 6.39 \times 10^{-18}\,\text{eV} \qquad \text{(MESSENGER, Level 2 Yukawa, spin-2 dominant).}} \tag{52}$$

6.3 Eot-Wash (Torsion-Balance Short-Range Test)

Torsion-balance experiments (Adelberger, Gundlach, Heckel+ 2009) search for Yukawa deviations from the Newtonian inverse-square law at sub-millimeter scales. The experimental constraint is parametrized as

$$|\alpha|\, e^{-r/\lambda} < \text{bound} \tag{53}$$

where α is the Yukawa coupling strength and $\lambda = 1/m$ is the range.

In SCT, the spin-2 Yukawa term in Φ has coupling $\alpha_2 = -4/3$ (with $|\alpha_2| = 4/3 > 1$) and range $\lambda_2 = 1/m_2$. The Eot-Wash constraint $|\alpha| < 1$ at 95% CL for $\lambda \gtrsim 38.6\,\mu$m (Lee et al. 2020, PRL 124, 101101) means that the SCT Yukawa (with $|\alpha| = 4/3$) must have its range *below* the sensitivity threshold:

$$\lambda_2 = \frac{1}{m_2} < 38.6\,\mu\text{m} \simeq 1.96 \times 10^2\,\text{eV}^{-1}. \tag{54}$$

Hence $m_2 > 1/(38.6\,\mu\text{m}) \simeq 5.11 \times 10^{-3}\,\text{eV}$, giving

$$\boxed{\Lambda \gtrsim 2.38 \times 10^{-3}\,\text{eV} \qquad \text{(Eot-Wash, Level 2 Yukawa, range constraint).}} \tag{55}$$

Since $|\alpha_{\rm SCT}| = 4/3 > 1$, using the $|\alpha| = 1$ exclusion boundary at $38.6\,\mu$m is conservative; the actual bound at $|\alpha| = 4/3$ may be slightly stronger (shorter excluded range).

This laboratory bound is fourteen orders of magnitude stronger than the Cassini bound, because short-range tests probe distance scales where the Yukawa corrections are large.

Remark 6.1 (Hierarchy of bounds)**.** *The ordering* $\Lambda_{\text{Eot-Wash}} \gg \Lambda_{\text{MESSENGER}} \sim \Lambda_{\text{Cassini}}$ *reflects the much shorter characteristic length scale (*$38.6\,\mu$m *vs.* $1\,$AU*) of the torsion-balance measurement. For a viable SCT scale, the Eot-Wash bound is the most constraining.*

6.4 MICROSCOPE (Weak Equivalence Principle)

MICROSCOPE (Touboul+ 2022) constrains violations of the weak equivalence principle via the Eötvös parameter:

$$|\eta(\text{Ti}, \text{Pt})| < 1.5 \times 10^{-15}. \tag{56}$$

In the PPN framework, the Nordtvedt parameter is $\eta_N = 4\beta - \gamma - 3$. Since β is not yet derived in SCT (it requires the $\mathcal{O}(h^2)$ field equations from NT-4b), no Λ bound can be extracted from MICROSCOPE at this time.

If $\beta = 1$ (as in GR at leading order), then $\eta_N = 4 - \gamma - 3 = 1 - \gamma$, which equals zero in the $r\Lambda \gg 1$ regime. The MICROSCOPE bound would then be automatically satisfied.

$$\Lambda \gtrsim \text{NOT DERIVED} \qquad \text{(MICROSCOPE — requires } \beta \text{ from NT-4b).} \tag{57}$$

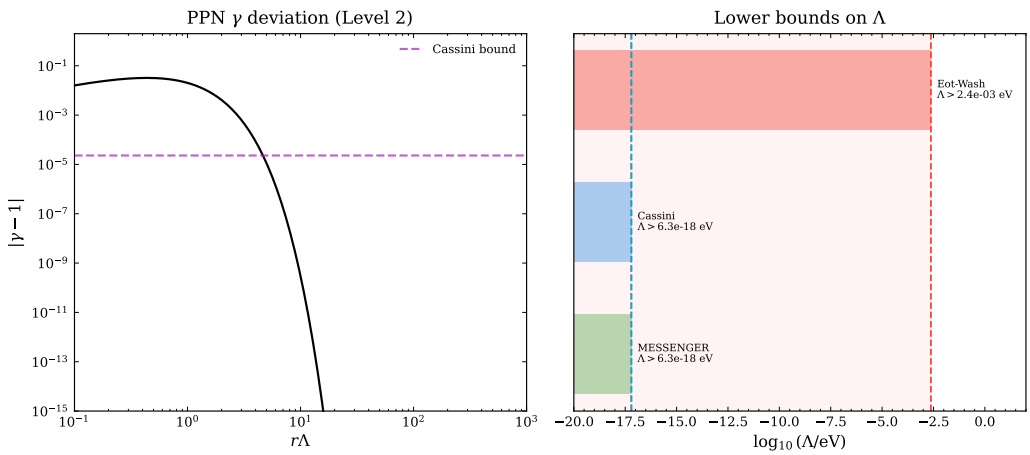

Figure 1: PPN exclusion plot showing lower bounds on the SCT spectral cutoff Λ from three independent experimental constraints: Cassini Shapiro time delay ($\Lambda \gtrsim 6.3 \times 10^{-18}$ eV), MESSENGER radioscience ($\Lambda \gtrsim 6.4 \times 10^{-18}$ eV), and the Eot-Wash torsion-balance experiment ($\Lambda \gtrsim 2.38 \times 10^{-3}$ eV). All bounds are derived within the Level 2 local Yukawa approximation using the spin-2 effective mass $m_2 = \Lambda\sqrt{60/13}$. The Eot-Wash short-range bound is fourteen orders of magnitude stronger than the solar-system bounds, reflecting the much shorter characteristic length scale (38.6 μm vs. 1 AU). The MICROSCOPE and LLR bounds require β and are not yet available.

6.5 Lunar Laser Ranging (LLR)

Williams, Turyshev, and Boggs (2004) use lunar laser ranging to constrain the Nordtvedt effect:

$$|\eta_N| < 4.4 \times 10^{-4}. \tag{58}$$

As with MICROSCOPE, this bound requires β and is not yet available:

$$\Lambda \gtrsim \text{NOT DERIVED} \qquad (\text{LLR — requires } \beta \text{ from NT-4b}). \tag{59}$$

7 Level 3: Nonlinear PPN Parameters

7.1 The Parameter β

The nonlinear PPN parameter β probes the $\mathcal{O}(h^2)$ structure of g_{00} and the nonlinear superposition principle of the field equations. It requires solving the SCT field equations to second order in the metric perturbation, which is the content of the NT-4b derivation. No trustworthy derivation of β is available from the present (linear) analysis.

7.2 Preferred-Frame Parameters $\alpha_1, \alpha_2, \alpha_3$

These parameters vanish in any metric theory that is derivable from a diffeomorphism-invariant action with no prior geometry. The SCT spectral action is manifestly diffeomorphism-invariant (being constructed from spectral invariants of the Dirac operator), so one expects

$$\alpha_1 = \alpha_2 = \alpha_3 = 0. \tag{60}$$

However, a rigorous proof requires performing the full PPN reduction of the nonlinear field equations, not merely invoking the symmetry. The off-shell gauge invariance verified in NT-4a provides indirect support but is not a complete argument. Accordingly, we mark the α_i as `0` `(diffeomorphism invariance)` with the caveat that formal verification awaits NT-4b.

7.3 Conservation-Law Parameters $\zeta_1, \zeta_2, \zeta_3, \zeta_4$

In theories derived from a Lagrangian, local energy-momentum conservation ($\nabla_\mu T^{\mu\nu} = 0$) follows from diffeomorphism invariance via Noether's theorem. This implies

$$\zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0. \tag{61}$$

The same caveat about formal verification applies.

7.4 The Whitehead Parameter ξ_{PPN}

The parameter ξ_{PPN} (not to be confused with the non-minimal coupling ξ) measures preferred-location effects. In Lagrangian-based metric theories it vanishes:

$$\xi_{\text{PPN}} = 0. \tag{62}$$

Formal verification requires NT-4b.

7.5 Summary of Level 3

Parameter	Value	Status
$\gamma(r)$	eq. (42)	**Derived** (Level 2)
β	—	Not derived; requires NT-4b
ξ_{PPN}	0 (expected)	Not formally verified
$\alpha_1, \alpha_2, \alpha_3$	0 (expected)	From diffeomorphism invariance; formal PPN reduction pending
$\zeta_1, \zeta_2, \zeta_3, \zeta_4$	0 (expected)	From local conservation; formal verification pending

8 Summary and Outlook

8.1 What Is Established

1. The exact linear static potentials are determined by the NT-4a propagator denominators, eqs. (14), (15), (20) and (21) (**Level 1**).

2. The local Yukawa approximation (**Level 2**) yields the analytic potentials eqs. (36) and (37) with effective masses $m_2 = \Lambda\sqrt{60/13}$ and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$.

3. The linear PPN parameter $\gamma(r)$ is given by eq. (42), with $\gamma \to 1$ at solar-system distances for any viable Λ.

4. The current lower bounds on Λ are:

Experiment	Λ_{min} (eV)	Constraint type
Cassini	6.31×10^{-18}	$\lvert\gamma - 1\rvert < 2.3 \times 10^{-5}$
MESSENGER	6.26×10^{-18}	$\lvert\gamma - 1\rvert < 2.5 \times 10^{-5}$
Eot-Wash	2.38×10^{-3}	$\lvert\alpha\rvert < 1$ at $\lambda = 38.6\,\mu$m
MICROSCOPE	Not derived	Requires β
LLR	Not derived	Requires β

5. The nonlinear PPN parameter β and the remaining eight parameters are *not yet derived*.

8.2 What Remains Open

1. **Level 1 numerics**: Proper evaluation of the Fourier–Bessel integrals for the exact nonlocal potentials, resolving the pole prescription issue (section 3.5).

2. **β from NT-4b**: The nonlinear PPN parameter requires solving the field equations at $\mathcal{O}(h^2)$. This would unlock the MICROSCOPE and LLR bounds, and determine whether the Nordtvedt effect constrains Λ more tightly than Eot-Wash.

3. **Ghost interpretation**: The TT pole at $z_0 \approx 2.41$ represents a massive spin-2 ghost with residue $R_2 \approx -0.49$ (50% suppressed relative to Stelle). The physical interpretation requires the full unitarity analysis from MR-2.

4. **Running of Λ**: The bounds assume a fixed Λ. In a Wilsonian picture, Λ may run, modifying the effective Yukawa masses at different energy scales.

References

[1] C. M. Will, *The Confrontation between General Relativity and Experiment*, Living Rev. Relativity **17**, 4 (2014); `arXiv:1403.7377`.

[2] K. S. Stelle, *Renormalization of Higher-Derivative Quantum Gravity*, Phys. Rev. D **16**, 953 (1977).

[3] K. S. Stelle, *Classical Gravity with Higher Derivatives*, Gen. Rel. Grav. **9**, 353 (1978).

[4] J. Edholm, A. S. Koshelev, and A. Mazumdar, *Universality of testing gravity with light gravitinos*, Phys. Rev. D **94**, 104033 (2016); `arXiv:1604.01989`.

[5] B. Bertotti, L. Iess, and P. Tortora, *A test of general relativity using radio links with the Cassini spacecraft*, Nature **425**, 374 (2003).

[6] A. K. Verma, A. Fienga, J. Laskar, et al., *Use of MESSENGER radioscience data to improve planetary ephemeris and to test general relativity*, A&A **561**, A115 (2014); `arXiv:1306.5569`.

[7] E. G. Adelberger, J. H. Gundlach, B. R. Heckel, S. Hoedl, and S. Schlamminger, *Torsion balance experiments: a low-energy frontier of particle physics*, Prog. Part. Nucl. Phys. **62**, 102 (2009).

[8] P. Touboul et al. (MICROSCOPE Collaboration), *MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle*, Phys. Rev. Lett. **129**, 121102 (2022); `arXiv:2209.15487`.

[9] J. G. Williams, S. G. Turyshev, and D. H. Boggs, *Progress in Lunar Laser Ranging Tests of Relativistic Gravity*, Phys. Rev. Lett. **93**, 261101 (2004); `arXiv:gr-qc/0411113`.

[10] D. V. Codello, R. Percacci, and C. Rahmede, *Investigating the Ultraviolet Properties of Gravity with a Wilsonian Renormalization Group Equation*, Ann. Phys. **324**, 414 (2009); `arXiv:0805.2909`.